

Understanding Transistor Aging Impact on the Behavior of RRAM Cells

Seyed Hossein Hashemi Shadmehri¹, Supriya Chakraborty¹, Thiago Santos Copetti¹,
Fabian Luis Vargas² and Letícia Maria Bolzani Poehls¹

¹ Chair of Integrated Digital Systems and Circuit Design, RWTH Aachen University, Germany

² IHP - Leibniz Institute for High Performance Microelectronics, Frankfurt (Oder), Germany

{hashemi, chakraborty, copetti, poehls}@ids.rwth-aachen.de, vargas@ihp-microelectronics.com

Abstract—Resistive Random Access Memories (RRAMs) have been considered for implementing various emerging applications alongside storage. RRAMs offer the opportunity to address the Von Neumann bottleneck making the implementation of In-Memory Computing (IMC) possible. However, the exploration of RRAM potentials depends on being able to guarantee their reliability during lifetime. In this context, this paper investigates the impact of transistor aging on the behavior of 1T1R RRAM cells. A case study composed of an RRAM block, including peripherals, implemented using the 22 nm FDSOI GF technology is adopted. The obtained results indicate that the transistor aging, which was previously widely ignored, can lead to parametric degradation or even catastrophic faults. In addition, the obtained results show that the transistor type used for implementing the 1T1R RRAM cell, core or IO device, and operating temperature play important roles when analysing aging impact.

Index Terms—Resistive Random Access Memory, Aging, Reliability

I. INTRODUCTION

Emerging applications, such as autonomous driving, image recognition and genomic data analysis, rely on large amounts of data. In a conventional Von Neumann computer, this data needs to be transferred between dedicated computation and memory units. Data movements are very costly in terms of delay and power consumption, leading to a severe loss of performance and energy efficiency [1]. In-Memory Computing (IMC) is a solution aimed at alleviating the Von Neumann bottleneck by integrating the computation inside the memory. There have been various attempts at realizing IMC, with some efforts augmenting the conventional memory types such as Flash with additional circuitry to embed processing capabilities [2]. Emerging Non-Volatile Memory (eNVM) technologies have also been heavily researched for their potential to enable IMC. Among them, Resistive Random Access Memories (RRAMs) have become one of the most promising candidates for this purpose. RRAMs can provide better latency and reduced energy consumption compared to Flash [3]. Furthermore, one can arrange RRAMs in a cross-point structure to facilitate $\mathcal{O}(1)$ execution of Vector Matrix Multiplication (VMM), which is at the core of neural network computations [4]. The most prevalent implementation of an RRAM cell is composed of one transistor and one memristive device (1T1R). In order to properly explore the potential of RRAMs, their reliability during lifetime needs to be assured. Reliability can be directly affected by temporal

variations, a type of time-dependent deviations, which degrade circuits during lifetime depending on the circuit's workload and operating conditions. Consequently, understanding how the aging of the transistor present in the RRAM cell may degrade the memory lifespan becomes essential. It is worth noting that previous studies have only focused on analysing the novel device's endurance itself, not considering the possible degradation of the transistor and the peripheral circuitry [5]–[9]. In this paper, an analysis of the possible transistor aging impact on the behavior of 1T1R RRAM cell is presented. The main contributions of this work are listed below:

- Development of an RRAM block using the 22 nm FD-SOI GF technology and the memristive device model described in [10].
- Comparison of the transistor aging impact on the RRAM cell when using IO and core devices as access transistors, across different temperatures.
- Analysis of the combined effect of transistor and memristive device aging on 1T1R RRAM cell.
- Exploration of body biasing as an approach for minimizing aging impact on 1T1R RRAM cell.

The rest of the paper is organized as follows: Section II offers a brief overview of failure mechanisms that can affect CMOS and memristive devices during lifetime. In Section III, the implemented case study and the adopted experimental setup are described. Section IV presents and discusses the obtained results and finally, Section V concludes this paper.

II. BACKGROUND

In this section, the failure mechanisms affecting CMOS and memristive devices during lifetime are summarized.

A. Transistor Degradation during Lifetime

The susceptibility of CMOS technology to temporal variations caused by phenomena, such as Hot Carrier Injection (HCI), Bias Temperature Instability (BTI) and Time-Dependent Dielectric Breakdown (TDDB), represents an important concern that becomes more prominent with technology scaling [11]. HCI occurs when charge carriers accelerated through channel area by the lateral field, gain enough energy to penetrate into the gate, creating traps at the gate-channel interface [12]. In general terms, BTI is caused by the stress induced by the vertical field in the presence of a zero to small

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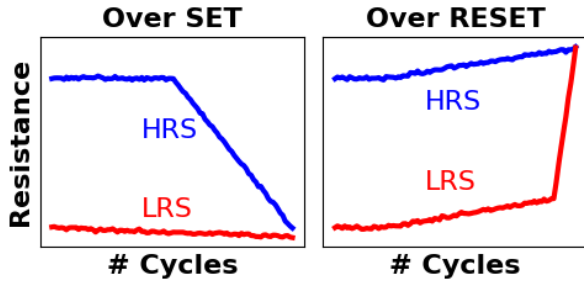


Fig. 1: Two types of RRAM endurance degradation [13].

lateral field. BTI can cause the Si-H bonds at the gate-channel interface to break and thus create charge traps. Depending on the considered BTI model, it is possible to assume that when the stress is removed, a recovery phase starts, decreasing the impact caused by this phenomenon. This recovery phase is not observed when considering the impact of HCI [12]. Finally, TDDDB is caused by the formation of a conductive path inside the gate dielectric, destroying its role as an insulator. BTI and HCI degrade the CMOS devices in a gradual manner, while TDDDB causes a sudden catastrophic failure [11]. It should be noted that in this paper BTI and HCI are the only degradation mechanisms being considered. BTI and HCI present similar behavior, being both caused by charge trapping in the gate dielectric, which causes the threshold voltage (V_{th}) to increase and carrier mobility to decrease, thereby reducing the current (I_{on}) and increasing the delay [11].

B. Memristive Device Degradation during Lifetime

In a Valence Change Mechanism (VCM)-based RRAM device, the formation and dissolution of a Conductive Filament (CF) governs the resistance. If oxygen ions are extracted from the intermediate layer and stored in the anode, the oxygen vacancies create a CF putting the device in Low Resistance State (LRS). If the stored ions are moved back and are recombined with the vacancies, the CF ruptures and the device will be in High Resistance State (HRS) [14]. The switching from HRS to LRS is called SET, and occurs when applying a V_{SET} voltage to the device. In contrast, the switching from LRS to HRS is called RESET, caused by applying a V_{RESET} voltage of opposite polarity to the device.

As observed in CMOS devices, RRAM devices are also susceptible to temporal variations that degrade their electrical parameters during lifetime. An analytical model for endurance degradation, based on measured data was proposed in [13]. This model includes two types of endurance degradation, as illustrated in Fig. 1. Over SET is the enlargement of the CF due to an excess in oxygen vacancy generation that leads to a drop in both HRS and LRS. Over RESET is the over-recombination of oxygen ions and vacancies, resulting in fewer oxygen vacancies, that cause an increase in both HRS and LRS. In both cases the conductance window shrinks and the ratio between HRS and LRS lowers significantly which is a reliability concern. The authors in [13], model endurance degradation by introducing a Δr and Δx into their RRAM

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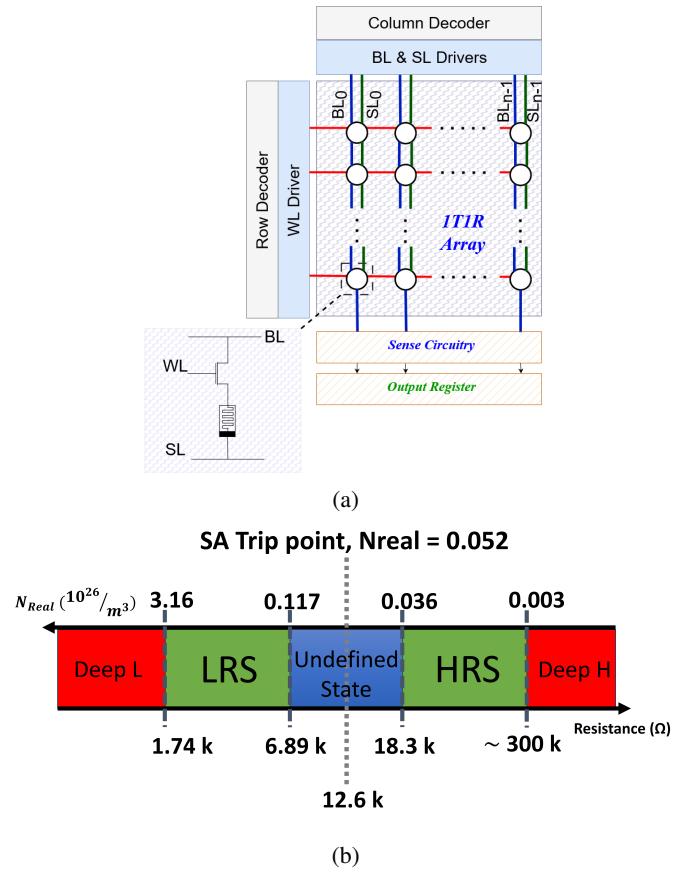


Fig. 2: (a) The RRAM block diagram and (b) N_{real} and corresponding resistive states: Deep L, LRS, Undefined State, HRS and Deep H.

model, representing the filament radius enlargement and reduction in minimum number of oxygen vacancies, respectively. It should be noted that Δr and Δx are applied at the same time and may affect the device in opposite ways.

III. EXPERIMENTAL SETUP

In order to understand the impact of temporal variations on the behavior of 1T1R RRAM cells, a case study implemented using the 22 nm FDSOI technology from GF and the memristive device model available in [10] was adopted. Fig. 2a depicts the case study's block diagram. SET operation is performed by driving the Bit Line (BL) to V_{set} and grounding the Select Line (SL), while RESET is performed by driving SL to V_{reset} and grounding the BL. During read, BL is driven to a small V_{read} voltage, while SL is grounded and the sensing circuitry makes a decision by comparing the bit cell's current with a reference current. Fig. 2b illustrates the boundaries adopted for representing the possible RRAM resistive states. In more detail, N_{real} , the number of oxygen vacancies per cubic meter in the disc region [10], and the respective resistance value for each possible resistive state (Deep L, Low Resistance State - LRS, Undefined State, High Resistance State - HRS and Deep H) are depicted. Higher N_{real} indicates a higher conductance (lower resistance).

TABLE I: RRAM block specifications

	Case Study 1	Case Study 2
Block size	8×8	8×8
Cell Type	1T1R with IO transistor	1T1R with core transistor
V_{WL}	2.0 V	1.2 V
V_{read}	0.4 V	0.4 V
SET	$V_{set} = 1.2$ V for $1\mu s$	$V_{set} = 1.2$ V for $50\mu s$
Body Bias	$V_{BS} = 0$ V	$V_{BS} = 800$ mV
RESET	$V_{reset} = 2.5$ V for $20\mu s$	$V_{reset} = 1.3$ V for $170\mu s$

It is important to highlight that two different 1T1R RRAM cells were implemented: The first one using IO devices and the second one using core logic devices. Table I summarizes the case study's specifications for the RRAM blocks implemented based on both device types. There is an obvious trade-off between the applied voltages during SET and RESET, and the time required for performing both operations. In many cases, this leads to the access transistor, 1T present in the RRAM cell, being over-driven so that a suitable operation speed is achieved [15]. It should be noted that the experiments in section IV-A and IV-B are performed by solely considering transistor degradation. However, the results summarized in section IV-C, combine the impact of transistor aging with memristive device degradation, allowing a more comprehensive analysis of 1T1R RRAM cell. In this work, Cadence Relxptool tool was used to simulate the effects of BTI and HCI. Furthermore, in this work, we use the RRAM model introduced in [10] named JART VCM v1b and according to [16]; the endurance failure observed for this device is stuck-at-LRS, meaning a degradation of the HRS state happens. Hence, we will focus on the over SET case. The parameter in JART model representing filament radius is r_{det} and thus increasing r_{det} creates the over SET (HRS degradation) effect.

IV. RESULTS AND DISCUSSION

This section summarizes the results obtained when considering the RRAM block implemented using IO and core logic devices as access transistors of the 1T1R RRAM cell. In addition, this section also analyses the aging impact on 1T1R RRAM cells, assuming the degradation of not only transistors, but also the memristive device.

A. IO Device Aging Impact

Considering bit cells implemented using IO devices, Fig. 3a shows the result for 5 and 10 years of aging caused by HCI and BTI at three different temperatures: -40°C , 25°C , and 125°C . The analysis was conducted by executing the operation sequence 0w1r1w0r0. Fig. 3b illustrates the effect of CMOS degradation on SET operation under different temperatures. The figure shows that at room temperature (25°C), aging increases the resistance achieved after SET by 0.45% after 5 and 0.57% after 10 years (with respect to fresh case). A similar behavior, but with less significance, can be observed when considering -40°C . In this case, the resistance increase after 5 years is around 0.11% and 0.14% after 10. Finally, the most noticeable impact occurs when considering a temperature of 125°C . In this case, the rise in resistance is 0.85% after 5 years and 1.08% after 10 years.

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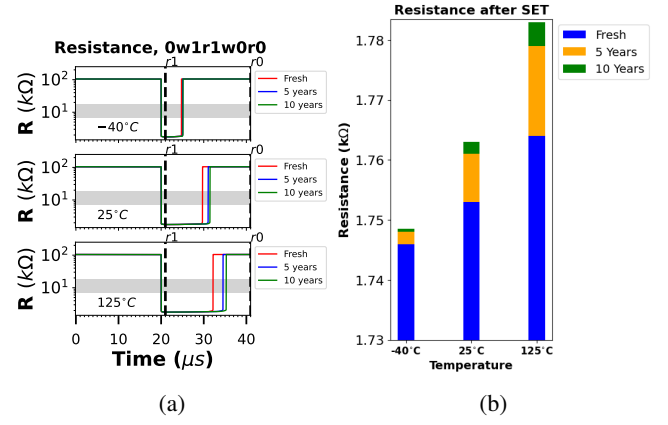


Fig. 3: Case study using IO devices (0w1r1w0r0): (a) Resistance vs Time (gray area shows region related to the Undefined State) (b) Resistance after w1 (SET operation).

TABLE II: Comparison of the RESET times (T_{reset}).

	-40°C		25°C		125°C	
	time	Δ	time	Δ	time	Δ
Fresh	$3.79\mu s$	-	$8.70\mu s$	-	$11.18\mu s$	-
5 years	$4.05\mu s$	6.9%	$10.02\mu s$	15.2%	$13.51\mu s$	20.8%
10 years	$4.12\mu s$	8.7%	$10.47\mu s$	20.3%	$14.27\mu s$	27.6%

In Fig. 3b it is observed that the resistance achieved after w1 operation was only slightly affected by the access device aging. However, the time required by the RESET operation (T_{reset}) was impacted more significantly. Table II summarizes the relation between access device aging and T_{reset} . In more detail, the access device aging increases T_{reset} and the effect is more pronounced at higher temperatures. For example, 10 years of aging would result in a 27.6% increase in T_{reset} at 125°C , while the figure for -40°C is only 8.7%.

B. Core Logic Device Aging Impact

The case study discussed in section IV-A was implemented using IO devices. A notable advantage of using IO devices is that they can tolerate much higher voltages before being affected by TDDDB. However, they have the obvious disadvantage of being much bigger than core logic devices and hence, greatly sacrificing density. Also, the forming process has been a hurdle in the way of core device utilization in RRAMs, since the process requires high voltages to be applied. However, in recent years, methods have been proposed for reducing the forming voltage. For example, the authors in [17] propose to reduce the forming voltage by either increasing the temperature or the rise time of the forming pulse, while [18] suggests using bipolar incremental step pulse programming. Developing such techniques has become a part of RRAM research. Additionally, works like [19] and [20] represent efforts towards demonstrating forming-free RRAMs. Also, the forming step is done only once and it lasts a short time. Furthermore, [21] by Infineon states that the lifetime of the core logic devices used as access transistors is a reliability concern because the voltages applied during SET/RESET are much greater than the permitted values for logic implementation. Thus, in this work we assume that the forming voltage is

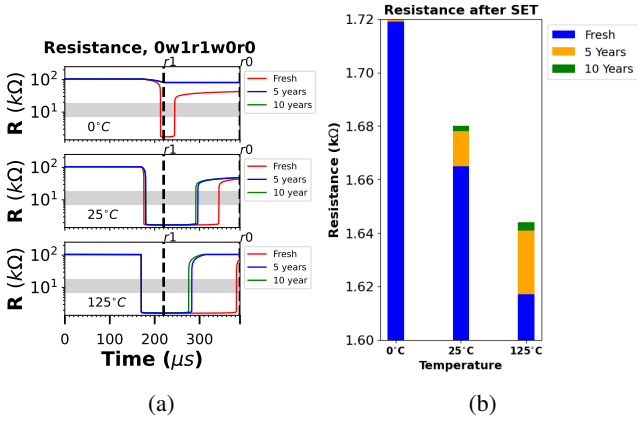


Fig. 4: RRAM block using core devices: (a) Resistance vs Time (gray area shows region related to the Undefined State), and (b) Resistance after w1, SET operation.

not going to be an issue for the core devices and focus on the post-forming operations (SET and RESET) of the 1T1R RRAM cell.

Fig. 4a illustrates the resistance of the 1T1R RRAM cell while performing the operation sequence 0w1r1w0r0, across three different temperatures. Considering the operation at 0°C, it is possible to observe a fault that will be propagated to the output of the block, since the RRAM cell was not able to properly switch. It should be noted that when using IO devices in the 1T1R cell, no functional fault was observed and transistor aging was merely able to cause parametric faults. Another important aspect to be considered is that the minimum considered temperature in case of core access device is 0°C, in contrast to -40°C that was used in section IV-A. The reason is that the SET operation has a positive feedback with temperature. In other words, the heat generated in the RRAM device by applying the SET pulse makes the SET process happen faster. Thus, SET operation starts more easily in high temperatures and is more difficult in low temperatures [10]. Hence, when adopting core devices to implement 1T1R cells, the SET operation in low temperatures becomes even more difficult and in this case it would be time prohibitive in sub-zero temperatures. SET and RESET can be divided into two steps, a slow conductance change step followed by a fast one. In case of SET, during the slow conductance change step, N_{real} rises (resistance drops) slowly and then during the fast conductance change step an abrupt rise in N_{real} occurs. The time needed for the first step of the process is called delay time and the time required for the second step is called transition time [10].

Fig. 4b shows the resistance of the target RRAM cell after the SET operation (after the end of w1 in the operation sequence). Comparing Fig. 4b with Fig. 3b, opposite trends in terms of resistance after the SET operation can be observed, when considering transistors with no aging (fresh). In the case of IO access device (Fig. 3b), the resistance achieved after SET increases with temperature (albeit slightly). In contrast, in case of 1T1R RRAM cells using core devices (Fig. 4b), the resistance achieved after SET decreases with temperature. This

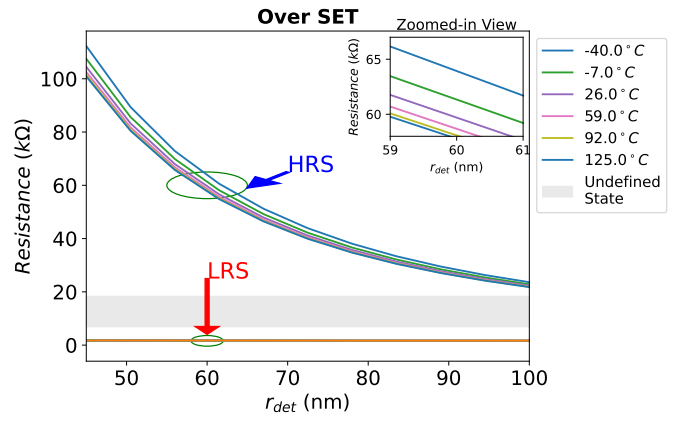


Fig. 5: Modeling over SET behavior by sweeping the r_{det} parameter from JART VCM v1b model [10].

can be attributed to the portion of SET time (T_{set}) belonging to delay time or transition time. For the case of IO device, the higher voltage used causes the delay time to be very small and most of the T_{set} being transition time. In case of core device, however, the device shows a noticeable delay time at 0°C but an insignificant one at 125°C. In other words, the temperature effect is more visible when employing core devices (low voltages). In case of IO device, there is an 18 Ω difference between the resistance after w1 at -40°C and 125°C when considering no aging. Meanwhile, the figure for core device is 102 Ω between 0°C and 125°C.

A further analysis of the results depicted in Fig. 4a and Fig. 3a, shows that the trends for T_{reset} also differ depending on the type of adopted transistors (IO or core device). In more detail, when using IO devices for implementing the 1T1R cell, the time required for RESET operation increases with aging. However, the opposite behavior is observed when using core devices. This occurs due to the fact that the switching dynamics of the RRAM is not only voltage or temperature dependant. The switching dynamics is also state dependant [10], meaning that the state of the RRAM cell (stored resistance) before performing the SET/RESET operation affects the outcome and speed of the operation. In more detail, Fig. 4a shows that the higher resistance of the RRAM cell after w1 in case of aged transistors, leads to lower T_{reset} and consequently, the RRAM cell with an aged transistor is able to go to HRS sooner.

C. Combined RRAM and Transistor Aging Impact

As discussed in section II-B, RRAM aging is characterized by a shrinkage of the conductance window. Furthermore, considering the adopted case study, the aging effect on the HRS is much more significant than that on the LRS. Hence we consider the case of over SET, which is modeled by sweeping the r_{det} variable (filament radius) of the JART model. Fig. 5 shows the effect of r_{det} on both HRS and LRS.

Once again we consider the 1T1R RRAM cell with the IO devices and the operation sequence 0w1r1w0r0. Fig. 6a shows the resistance of the RRAM device after SET and the time required for RESET operation (T_{reset}), at different temperatures and for various filament sizes, in the case of

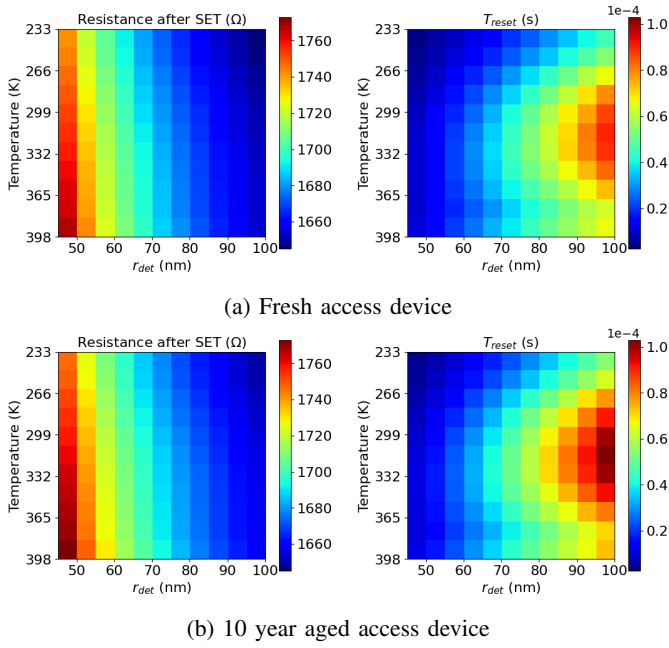


Fig. 6: RRAM block with IO devices: Resistance after SET and T_{reset} at different temperatures and various CF sizes when performing 0w1r1w0r0. (a) Fresh access device and (b) 10 year aged access device.

a fresh access device. In other words, in this case only the RRAM aging has been considered. It should be noted that filament radius was swept from the nominal value of 45 nm to the maximum value of 100 nm allowed by JART VCM v1b model [10]. It can be observed that having a bigger filament radius leads to achieving a lower resistance after the SET operation but at the same time it makes the following RESET operation more difficult. An increase in temperature leads to higher resistance after SET. From equal starting points, an increase in temperature also increases T_{reset} , however here we see the opposing effects of initial state (resistance after previous SET) and temperature. This is why in some cases we observe maximum (T_{reset}) at intermediate temperatures. Fig. 6a captures the interplay between filament radius (representing HRS degradation) and temperature change. A combination of high temperature and small filament radius is the worst case for the resistance after SET, while the combination of intermediate temperatures (42°C here) and big filament radius is the worst case for (T_{reset}).

Fig. 6b considers the case which the access device has experienced 10 years of aging. It can be seen that while the trends discussed above for r_{det} and temperature remain true, there is an overall increase in both the resistance after SET and the required time for the following RESET operation. This suggests that only considering RRAM aging in studying the reliability of the memory blocks, could lead to an underestimation of the aging effects.

D. Exploring Body Biasing

FDSOI is especially suitable for body biasing schemes thanks to the Buried Oxide (BOX) isolating the channel from

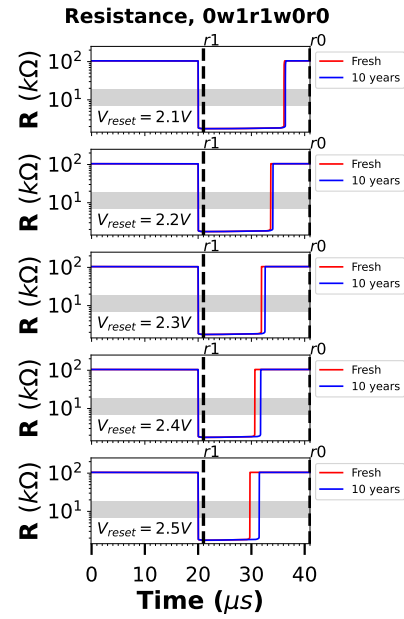


Fig. 7: RRAM block with IO devices: Sweeping the V_{reset} , while executing 0w1r1w0r0.

the substrate, allowing the application of a wider range of body bias [22]. Considering the case study implemented using IO devices, Fig. 7 illustrates the effect of sweeping V_{reset} , while executing the operation sequence 0w1r1w0r0 at room temperature (25°C). It can be observed that decreasing V_{reset} reduces the difference between behavior of the fresh and aged RRAM cells. This behavior is expected, since the rates of mobility and V_{th} degradation due to BTI and HCI depend on the stress conditions. However, this robustness against aging has come at the price of reducing performance, such that there is an increase of 21.65% in the T_{reset} , when comparing the case study assuming a $V_{reset} = 2.5$ V with one assuming $V_{reset} = 2.1$ V.

Fig. 8 shows the comparison between the original RRAM block using IO devices (see specification characteristics in Table I) and the new RRAM block adopting $V_{reset} = 2.1$ V with forward body biasing of 250 mV during w0. Observing Fig. 8, it is visible that body biasing has elevated the performance with $V_{reset} = 2.1$ V, such that it's T_{reset} almost matches that of $V_{reset} = 2.5$ V with no body bias. It should also be noted that body bias was only applied during w0, because it's application during w1 would have lead to gathering more oxygen vacancies in the disc region (N_{real}), which in turn makes RESET more difficult. Table III summarizes the information regarding the application of body biasing.

V. CONCLUSION

In this paper, an analysis of the transistor aging impact on the behavior of 1T1R RRAM cells was presented. Two different case studies, one considering IO devices and another one considering core logic devices for implementing the access transistors of RRAM cells, were developed. The obtained results demonstrated that despite providing better density, core logic devices show slower operation and greater reliability

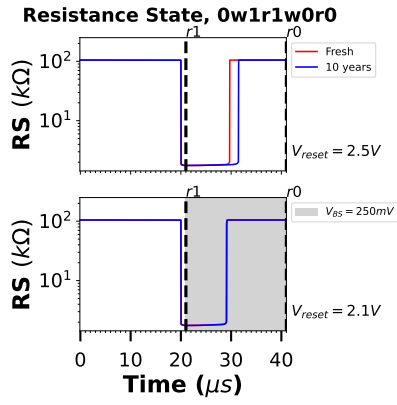


Fig. 8: RRAM block with IO devices, Top: Case Study 1 (Table I) vs. Bottom: Exploring body biasing and reduced voltage levels.

TABLE III: Comparison of the RESET times (T_{reset}).

	Fresh	10 years	Δ
$V_{\text{reset}} = 2.5V$ & $V_{BS} = 0V$	$8.7\mu s$	$10.47\mu s$	20.3%
$V_{\text{reset}} = 2.1V$ & $V_{BS} = 0V$	$15.12\mu s$	$15.37\mu s$	1.65%
$V_{\text{reset}} = 2.1V$ & $V_{BS} = 0 - 250mV$	$8.05\mu s$	$8.12\mu s$	0.87%

issues. In addition, the performed simulations show greater degradation in performance when considering combined effects of transistor and memristive device aging, compared to considering only RRAM endurance degradation. Preliminary results show that body biasing can be explored as an alternative for reducing the stress on the access transistor of the 1T1R RRAM cells, increasing cell's robustness against temporal variations.

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